

Five methodological refusals. Each names a default the field accepts and that this dissertation chose to refuse.

What I refused to do

Report trial-level cross-validation (typically inflates clinical-EEG accuracy by 10-15 percentage points by leaking same-subject correlation across train and test).

Use parametric p-values that assume noise-structure models EEG channel selection does not satisfy.

Select β by maximizing validation accuracy.

Treat autonomous $p(W) < 1$ as sufficient evidence that the reservoir is in the usable regime.

Justify the reservoir architecturally through Maass-Markram alone (a non-constructive existence theorem).

What I did instead, and why

Subject-level StratifiedGroupKFold across every classification claim in the dissertation. Trial-level numbers were never the headline.

→ Chapter 5; Exp D

Per-trial channel-permutation null (Exp D) that assumes nothing about channels and tests the spatial claim the dissertation actually makes.

→ Exp D

Select β within the measured memory-capacity plateau (Exp C), anchored to a biological time-constant argument ($1/\beta \approx 20$ steps matches the α -period). The choice does not move when the downstream task changes.

→ Exp C

Measure driven λ_1 along 3,165 real ERP trajectories (Exp A). The reservoir is in the usable regime BECAUSE we measured it there.

→ Exp A

Measure the data's Takens dimension and verify the reservoir's state space exceeds it by 16-64 $\times$ (Exp B). The justification is data-specific, not generic.

→ Exp B

Each refusal is a rigor commitment. Each commitment is in the experiments above.